GITAM Deemed to be University

Computer Science and Engineering | Course Code: CS101

Object-Oriented Programming & Principles - Comprehensive Syllabus Lectur

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Syllabus Focus: CS101 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Object-Oriented Programming & Principles. Inc

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Paradigms & Object-Oriented Fundamentals

" Syllabus Scope: Procedural vs Object-Oriented paradigms, Data encapsulation, Access specifiers, Constru

Unit 2: Inheritance & Dynamic Binding

" Syllabus Scope: Single, multiple, and hierarchical inheritance, Diamond problem, Virtual base classes,

CS101: Object-Oriented Programming & Principles - Advanced Topics & Assessment

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Memory Architecture & RAII

" Stack vs Heap memory allocation, RAII idiom, Smart pointers unique_ptr, shared_ptr, weak_ptr, Generic templa

Unit 4: Exception Hierarchies & Stream I/O

" Try-catch-throw mechanisms, Custom exception classes, File streams, Serialization, Binary and formatted disk

Unit 5: Modern Language Idioms & Design

" Lambda expressions, Move semantics, Rvalue references, Operator overloading, Thread concurrency primitives

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Illustrate with memory diagrams how virtual method tables vtable and virtual pointers vptr achieve run
2. Describe the diamond problem in multiple inheritance and show how virtual base classes resolve duplica
3. Explain RAII Resource Acquisition Is Initialization and contrast unique_ptr with shared_ptr in referen

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Bjarne Stroustrup - The C++ Programming Language Addison-Wesley 4th Ed
* Robert Lafore - Object-Oriented Programming in C++ Sams Publishing

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